

Carter O.B. Bliss

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Objective

Test automation and embedded systems engineer with hands-on experience across firmware, PCB design, and hardware validation, seeking a Summer 2027 internship in systems, hardware, or software engineering.

Education

Georgia Institute of Technology | B.S. Electrical Engineering | E.G: 2028 | GPA: 4.00

- Concentration: Electronic Devices, Robotics Autonomous Systems
- Minor: Computer Science – Intelligence

EXPERIENCE

Systems Engineering Intern – L3Harris Technologies

Jun 2026 – Aug 2026

- Engineered autonomous color calibration software in NI TestStand for the MCDU-R Airbus and B777 DLO table, architected for portability across additional unit types (e.g., ECMDU, CMDU).
- Interfaced system with CS3000, Westar FPM H, and CS200 equipment, executing factory test commands and implementing a CS200 driver in Python for integration to acceptance test procedures.
- Reduced calibration time from **80 hours to 15 minutes**, an estimated **\$11,200 in annual savings** across MCDU-R production pipeline.
- Led design verification testing for AMU CNBP, CMFD, and MCDU-R units: exposed graphics functional failures, eliminated over 100 instances of procedural ambiguity, and authored new color validation procedure.

Electrical & Firmware Engineer – HyTech Racing

Aug 2025 - Present

- Developed C++ steering sensor firmware on Teensy 4.1 to calibrate, run plausibility checks, and convert ADC values for analog and digital steering sensor; validated with Google Test unit tests across **25+ edge cases**.
- Integrated CAN & Ethernet data transmission to Drivebrain, enabling “Mode 4” motor vehicle dynamics: real-time software adjustments with active torque compensation for high-speed cornering.
- Enabled HTX car to reduce skid-pad time by **~0.2 seconds** due to feedback from steering system into Drivebrain at Formula South 2026.
- Created an FSAE-compliant brake light PCB meeting competition timing standards for team’s electric racecar.

Advanced Prototyping Technical Officer – Materials Innovation Learning Laboratory

Aug 2025 - Present

- Manage, maintain, and troubleshoot material testing equipment to assist undergraduate and graduate students.
- Create standard operating procedures and lead hands-on staffer trainings in Advanced Prototyping Lab.
- Provide hands-on technical assistance to approximately **15 users and staffers weekly**, promoting safe equipment usage and experimental workflows.

Systems Designer – Georgia Tech Vertically Integrated Projects

Jan 2026 – May 2026

- Led a team of 4 as experiment card circuit design lead on an interdisciplinary project characterizing team-fabricated supercapacitors on a NASA stratospheric balloon flight.
- Designed and simulated (KiCad, LTspice) constant-current charge/discharge subcircuits, custom op-amp feedback loops, PMOS/NMOS switching networks for timed cycling, and overvoltage protection circuitry.
- Developed Rust firmware for an STM32 microcontroller to drive PWM switching and synchronize ADC current/voltage sampling.

Projects

Steering Firmware & Brake Light PCB – HyTech Racing

Fall 2025

- Designed brake light PCB in Altium Designer: created custom LM555 8-SOIC schematic symbol and footprint from datasheet, laid out RC timing circuit with USB-C power input, completed full DRC with 0 errors, and assembled board.
- Implemented steering sensor firmware within HyTech’s VCF codebase in C++; wrote unit test suite using Google Test covering calibration, plausibility checks, and ADC conversion cases.

Skills & Hobbies

- Programming: Python, C#, C++, C, MATLAB
- Software: NI TestStand, Altium, KiCad, LTspice, Blender, Excel, GitHub, Power BI, Glowscript, Tinkercad
- Hardware: Teensy 4.1, STM32
- Lab Equipment: Prusa i3 & Formlabs 3+ printers, F505 Mechanical Tester, Glowforge Laser Cutter, CNC
- Languages: English(native), Mandarin(proficient)
- Leadership: DVHS XC & Track Captain, BSA Patrol Leader, USNA SS, Philmont - Head Chef
- Hobbies: Weight lifting, travelling, running, ping pong, learning guitar